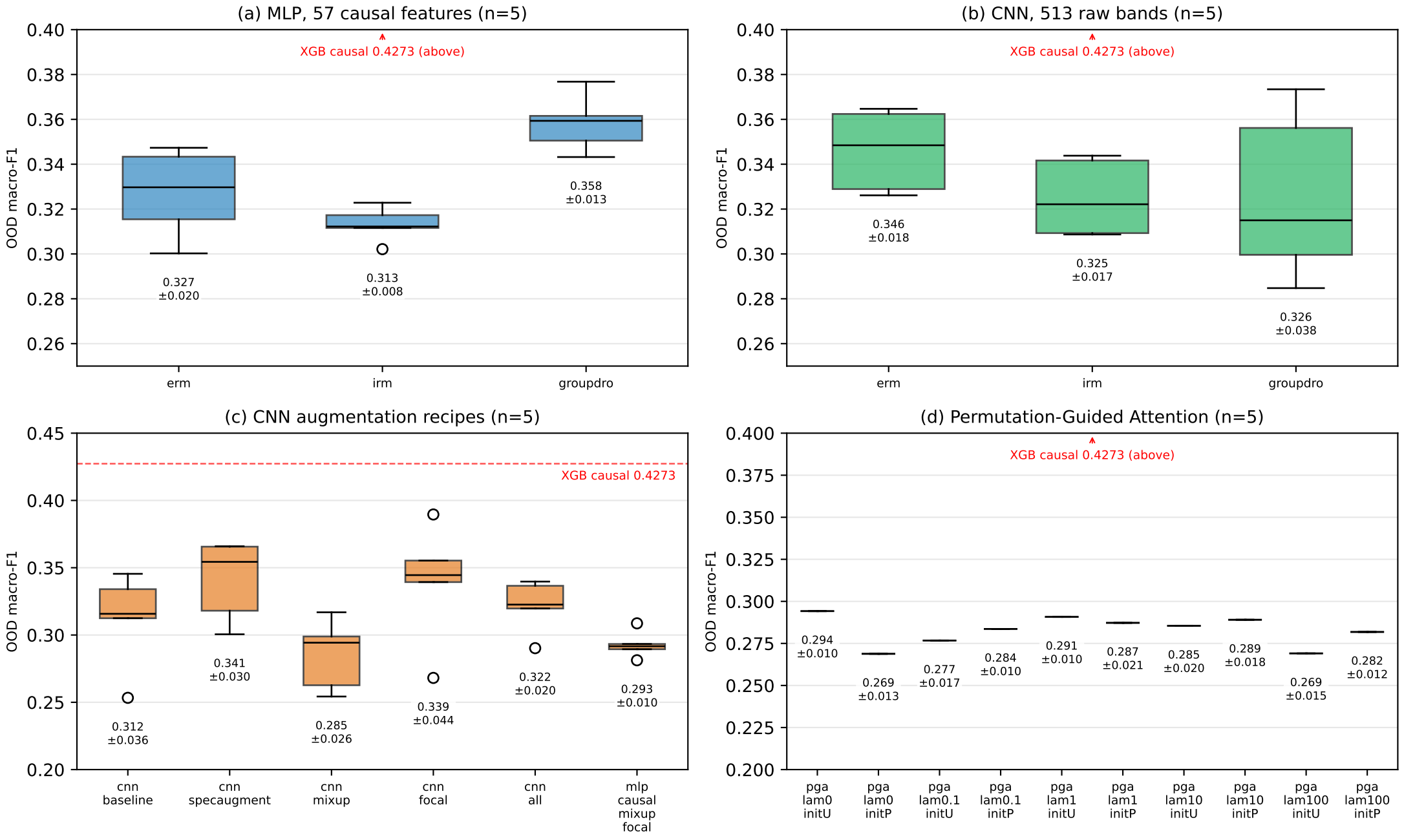
(a) MLP, 57 causal features (n=5)
XGB causal 0.4273 (above)
OOD macro-F1
0.40
0.38
0.36
0.34
0.32
0.30
0.28
0.26
0.327
±0.020
0.313
±0.008
0.358
±0.013
erm
irm
groupdro
(b) CNN, 513 raw bands (n=5)
XGB causal 0.4273 (above)
OOD macro-F1
0.346
±0.018
0.325
±0.017
0.326
±0.038
erm
irm
groupdro
(c) CNN augmentation recipes (n=5)
XGB causal 0.4273
OOD macro-F1
0.45
0.40
0.35
0.30
0.25
0.20
0.312
±0.036
0.341
±0.030
0.285
±0.026
0.339
±0.044
0.322
±0.020
0.293
±0.010
cnn baseline
cnn specaugment
cnn mixup
cnn focal
cnn all
mlp causal mixup focal
(d) Permutation-Guided Attention (n=5)
XGB causal 0.4273 (above)
OOD macro-F1
0.400
0.375
0.350
0.325
0.300
0.275
0.250
0.225
0.200
0.294
±0.010
0.269
±0.013
0.277
±0.017
0.284
±0.010
0.291
±0.010
0.287
±0.021
0.285
±0.020
0.289
±0.018
0.269
±0.015
0.282
±0.012
pga lam0 initU
pga lam0 initP
pga lam0.1 initU
pga lam0.1 initP
pga lam1 initU
pga lam1 initP
pga lam10 initU
pga lam10 initP
pga lam100 initU
pga lam100 initP